

Jisung Woo

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https://jisungwoo.github.io/jisung_website_v5/

DATA ENGINEER

AI-focused Data Engineer with 5+ years owning enterprise data delivery end-to-end — from stakeholder requirements through offshore team delivery, production deployment, and hyper care — on a PB-scale Snowflake EDW under government compliance requirements. Builds with AI as a platform rather than a tool: driving LLM-based documentation automation at Honeywell and shipping MelodyMap.ai, an AI-orchestrated full-stack SaaS, as a solo founder.

EXPERIENCE

Data Engineer II

August 2024 – Current

Honeywell Aerospace

Phoenix, AZ

- Own end-to-end delivery of EDW data products on a PB-scale Snowflake warehouse serving 100K+ employees — gathering requirements from directors, translating them into development specs, and directing 10+ offshore engineers through code review, QA on restricted government data, and production deployment.
- Serve as the primary technical liaison between business and engineering — communicating directly with directors on scope, timelines, and data quality, and converting evolving business requirements into SQL transformation logic across 10M+ row government datasets under strict compliance requirements.
- Cut runtime of a critical transformation pipeline by 3x (1.5 hours → 30 minutes) after a major business-logic expansion, by re-architecting a monolithic design view into a staged pattern (design view → staging table → design view → data model) with zero change to output data.
- Developing an automated mapping-documentation pipeline using Snowflake's native AI (Cortex) that parses complex design-view SQL — often 100–300 columns per view — and generates business- and security-facing mapping documentation, plan to reduce at least 3 hours of manual extraction per view previously done by offshore engineers.
- Architected and prototyped an internal React-based data lineage visualization tool delivering Atlan-style interactive flow diagrams, demonstrating the feasibility of in-house data catalog tooling to enterprise leadership; leading design of the EDW's semantic layer and table/column lineage framework.

Data Engineer I

March 2022 – July 2024

ASU Enterprise Partners

Tempe, AZ

- Automated and migrated ETL/ELT pipelines using SSIS, Azure Data Factory, and SQL Server, integrating data from Salesforce, Workday, and Azure Blob Storage into the enterprise data warehouse.
- Developed SQL Server stored procedures to automate Power BI dashboard updates and implemented data deduplication routines, improving reporting accuracy and reducing DAX complexity.

Technical Solution Analyst

February 2021 – March 2022

Oracle - Cerner Corporation

Kansas City, MO

- Investigated and resolved complex front-end and backend data issues across enterprise applications using SQL, log analysis, and proactive Oracle-based monitoring—maintaining 90%+ client satisfaction while meeting SLAs.

PROJECT

Founder / Full-Stack AI Product Engineer

April 2026 - Current

Personal Project - MelodyMap.ai

- Designed and built a full-stack, HIPAA-adjacent SaaS end-to-end (Next.js/TypeScript, FastAPI/Python, PostgreSQL + pgvector, LLM integration via a backend-only model gateway) that automates the 1–5 hours of daily unpaid planning music therapists do by hand – session plans, clinical reports, and mileage logs.
- Architected a hybrid AI system: a deterministic recommendation engine (vector similarity + domain signals such as population profiles, therapist repertoire, and prior session outcomes) serves as the source of truth, with an LLM composing explanations around it — preventing hallucinated clinical facts.
- Engineered a PHI-minimization pipeline keeping protected health information out of all LLM calls, plus a self-improving memory loop where post-session debriefs feed future recommendations.
- Built the entire application AI-first — directing AI coding agents as the execution engine, with an independent AI reviewer gating changes through a 3-environment CI/CD pipeline (Dev → QA → Prod) with automated dependency and security scanning, including OWASP Top 10 audits.

SKILLS

- **AI & LLM:** Multi-Agent Orchestration, LLM-Assisted Development, Prompt & Context Engineering, Vector Search (pgvector), Snowflake Cortex, Claude, Codex, GitHub Copilot, Ollama
- **Languages:** SQL (expert), Python, TypeScript/JavaScript, HTML/CSS, Bash
- **Data Engineering:** Snowflake, Informatica, Control-M, SSIS, SQL Server, ETL/ELT, Data Modeling, Data Warehousing, Power BI
- **Platforms & Tools:** Azure, FastAPI, Next.js, React, PostgreSQL, Supabase, GitHub, CI/CD, Salesforce
- **Spoken Languages:** English (Native), Korean (Native)

EDUCATION

Arizona State University, W. P. Carey School of Business – M.S. in Business Analytics

August 2023 – December 2024

Arizona State University – B.S. in Computer Information Systems

August 2016 – December 2020